

[129]

SARDAR PATEL UNIVERSITY

BCA Semester-6 External Examination [NC] – April-2026

US06CBCA23 || Data Communication & Computer Networks [Old Course]

Date : 04-04-2026

Time : 10.00 a.m. – 01.00 p.m.



Total : 70 Marks

Q-1 **Select the appropriate option for the following questions:**

10

- 1) Traditional LANs run at speeds of _____ to _____ Mbps.
a) 20 to 100
b) 10 to 1000
c) 100 to 1000
d) 10 to 100
- 2) The connection of two or more networks is called an _____.
a) computer network
b) internet
c) internetwork
d) local Area Network
- 3) A _____ cable is made of glass or plastic and transmits signals in the form of light.
a) Twisted-pair
b) Coaxial cables
c) Fiber Optical
d) None of these
- 4) _____ is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.
a). Multiplexing
b). Demultiplexing
c). CDMA
d).None of these.
- 5) A set of layers and protocols is called _____.
a) network architecturer
b) Protocol stack
c) Computer architecture
d) Layers stack
- 6) Routing is the responsibilities of _____.
a) Network layer
b) Session layer
c) Transport layer
d) Application layer
- 7) Which of the following topology requires maximum cabling?
a) Bus
b) Star
c) Ring
d) Mesh
- 8) A repeater operates at _____ layer.
a) physical
b) datalink
c) network
d) all of these
- 9) Mail and directory service is responsibility of _____.
a) Presentation layer
b) Application layer
c) Network layer
d) Session layer
- 10) "Interconnected collection of autonomous computers" is known as _____.
a) Computer network
b) Host
c) Subnet
d) Server

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE)

20

- 1) Define terms : Topology, Data Rate
- 2) What is computer network? What are the three categories of computer network?
- 3) Explain DQDB in brief.

- 4) Explain Magnetic media in short.
- 5) Differentiate between TDM and FDM.
- 6) What do you mean by Multiplexing? List out various Techniques.
- 7) List categories of satellite.
- 8) Explain responsibilities of Physical layer.
- 9) Explain orbit.
- 10) Write full form of CSMA. List different types of CSMA.
- 11) List various topologies.
- 12) List Network devices.

- Q-3** A What are the uses/advantages & Disadvantages of computer networks 5
B Explain LAN in detail. 5

OR

- Q-3** A Explain classification of computer Network. 5
B Explain MAN in detail. 5

- Q-4** A Explain Co-axial cable in detail. 5
B Explain Twisted pair cable in detail. 5

OR

- Q-4** A Explain serial transmission in detail. 5
B Explain Parallel transmission in detail. 5

- Q-5** Draw diagram and explain all layers of OSI model. 10

OR

- Q-5** Write a note on protocol hierarchies & design issues for the layers. 10

- Q-6** A Explain Star topology in detail. 5
B Explain Ring topology in detail. 5

OR

- Q-6** A Explain Repeater in detail. 5
B Explain Bridge in detail. 5

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